

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 12: RNAV Architecture, FMS Sensors & Total System Error

Syllabus Source: CAE Oxford (pp. 473-520) | Pass Target: $\geq 90\%$ AviationExam

1. FMS MULTI-SENSOR POSITION UPDATING HIERARCHY

Modern Flight Management Systems (FMS) compute a continuous best-estimate position by fusing inputs from multiple sensors:

- 1. GNSS (GPS / Galileo): Primary sensor offering highest continuous accuracy (< 5 m).
- 2. DME / DME Updating: Secondary terrestrial sensor. Requires at least two DME stations whose intersecting radials cross at an angle between 30° and 150° (optimal 90°)! Sub-mile accuracy.
- 3. VOR / DME Updating: Single VOR/DME co-located station providing bearing and distance. Lower accuracy due to VOR angular divergence at range.
- 4. Inertial Reference Systems (IRS): Provides continuous dead-reckoning position during GNSS or DME reception loss. IRS position drifts over time.

2. TOTAL SYSTEM ERROR (TSE) COMPONENTS

Under ICAO PBN definitions, the lateral error of an aircraft relative to the intended track consists of three statistically independent errors combined using root-sum-square:

- The TSE Master Formula: $TSE = \sqrt{PDE^2 + FTE^2 + NSE^2}$
- Path Definition Error (PDE): Discrepancy between the defined path in the navigation database and the intended true flight path (virtually zero in modern digital databases).
- Flight Technical Error (FTE): Pilot or autopilot accuracy in tracking the indicated path on the flight director/CDI.
- Navigation System Error (NSE): The difference between actual aircraft position and the position computed by the navigation sensors.

3. FLY-BY VS FLY-OVER WAYPOINTS

- Fly-By Waypoint (Standard): Depicted as an open 4-point star. The FMS initiates turn anticipation BEFORE reaching the waypoint, cutting the corner smoothly without overshoot.
- Fly-Over Waypoint: Depicted as a 4-point star enclosed in a circle. The aircraft MUST FLY DIRECTLY OVERHEAD the waypoint before initiating the turn, followed by an overshoot and intercept.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Total System Error (TSE) Calculation

Mathematical Formula: - $TSE = \sqrt{PDE^2 + FTE^2 + NSE^2}$. Example: - Path Definition Error (PDE) = 0.05 NM (database error, usually negligible). - Navigation System Error (NSE) = 0.20 NM (sensor error). - Flight Technical Error (FTE) = 0.15 NM (autopilot tracking error). Step 1: Sum of squares = $0.20^2 + 0.15^2 = 0.04 + 0.0225 = 0.0625$. Step 2: $TSE = \sqrt{0.0625} = 0.25$ NM. Conclusion: Since TSE (0.25 NM) < 0.3 NM, the aircraft meets RNP APCH certification standards!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Total System Error (TSE)

SCENARIO (EASA PBN Total System Error Drill): During an RNP 1 terminal departure, the FMS computes the error components: - Path Definition Error (PDE) = 0.10 NM - Flight Technical Error (FTE) = 0.50 NM - Navigation System Error (NSE) = 0.60 NM QUESTION: What is the Total System Error (TSE), and is it compliant with the RNP 1 specification (+/- 1.0 NM)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the Master TSE Root-Sum-Square Formula: - Formula: $TSE = \sqrt{PDE^2 + FTE^2 + NSE^2}$ Step 2: Square each error component: - $PDE^2 = 0.10^2 = 0.01$ - $FTE^2 = 0.50^2 = 0.25$ - $NSE^2 = 0.60^2 = 0.36$ Step 3: Sum the squared values: - $Sum = 0.01 + 0.25 + 0.36 = 0.62$ Step 4: Take the square root: - $TSE = \sqrt{0.62} = 0.787 \text{ NM} (\sim 0.79 \text{ NM})$ Step 5: Compare with RNP 1 Containment Limit (1.0 NM): - $0.79 \text{ NM} \leq 1.0 \text{ NM} \rightarrow$ COMPLIANT! FINAL ANSWER: Total System Error is 0.79 NM, which is well within the 1.0 NM legal limit for RNP 1.

[!] EXAM TRAP (AVIATIONEXAM) - The DME/DME Intersection Angle Trap

- For an FMS to accept a DME/DME position update: - The angle between the two DME station lines of position MUST be between 30° and 150°! - Angles under 30° or over 150° create severe geometric dilution and are automatically rejected.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What are the acceptable intersection angles between two DME stations for an FMS DME/DME position update?

- [A]: Exactly 90° only
- [B]: BETWEEN 30 DEGREES AND 150 DEGREES (with optimal geometry at 90°)
- [C]: Any angle
- [D]: Less than 15°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The FMS rejects DME pairs intersecting at angles shallower than 30° or wider than 150° to avoid geometric position stretching.

Q2: In Performance-Based Navigation, what are the three components of TOTAL SYSTEM ERROR (TSE)?

- [A]: Wind error, drift error, clock error
- [B]: PATH DEFINITION ERROR (PDE), FLIGHT TECHNICAL ERROR (FTE), and NAVIGATION SYSTEM ERROR (NSE)
- [C]: Altimeter error, speed error, heading error
- [D]: Magnetic error only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By ICAO definition, $TSE = \sqrt{PDE^2 + FTE^2 + NSE^2}$, combining database, pilot/autopilot tracking, and sensor errors.

Q3: What symbol on an instrument flight chart depicts a 'FLY-OVER' waypoint?

- [A]: A plain 4-point star
- [B]: A 4-point star ENCLOSED WITHIN A CIRCLE

- [C]: A solid black triangle
- [D]: An open square

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A fly-by waypoint is an unenclosed star; enclosing the star in a circle mandates flying directly overhead before turning.